

Diagnostic Evaluation Report

1. Framework Information

- **Framework:** X-RAG Diagnostic Framework (v1.0)
- **Trace ID:** `8eb23758-4b1e-41d3-810c-2367ffbdec56`
- **Timestamp:** 2026-09-05T16:08:23.677710Z
- **Status:** COMPLETED

2. Executive Summary

- **Question:** Which judgments in this corpus apply or construe Section 235?
- **Answer:** The retrieved corpus contains the **text** of Section 235 of the Bharatiya Nyaya Sanhita, 2023 (see, for example, the provisions quoted in chunks [2390f58b-03a1-4c95-b465-1c314bc6447a] and [89e6be59-9cb1-473a-b635-4a7478c6c5aa]), but it does **not** include any judicial 235.
- **Health Score:** 0.88
- **Primary Issue:** UNKNOWN
- **Claims Supported:** 5 / 5
- **Summary:** Analysis completed.

3. Pipeline Overview

Stage	Status	Confidence	Observation
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CORPUS	PASS	0.9	Corpus contains sufficient relevant content, evidenced by at least one supported claim.
RETRIEVER	PASS	0.95	At least one retrieved chunk provides supporting evidence for a verified claim.
GENERATOR	PASS	0.95	All generated claims are fully supported by evidence.
GROUNDING	PASS	0.9	No claims are contradicted by retrieved evidence.

Efficiency: 1/6 chunks used (17%)

4. Evaluation Metrics

- **Retrieved Chunks:** 6
- **Verified Claims:** 5 (Supported: 5, Partial: 0, Unsupported: 0, Contradicted: 0)
- **Grounding Score:** 1.00
- **Evidence Coverage:** 0.17
- **Average Entailment:** 0.86
- **Latencies:** Retrieval (951.9ms) | Generation (13610.7ms) | Verification (8744.5ms)

5. RAGAS Metrics

- **Faithfulness:** 1.0000
- **Answer Relevancy:** 0.8257
- **Context Precision:** 0.6389
- **Context Relevancy:** 0.0000
- **Context Recall:** N/A
- **Answer Similarity:** N/A
- **Answer Correctness:** N/A

Answer Correctness (Claim Recall)

Not computed for this trace.

6. Evidence Analysis

- **[22a6bbe7-9c32-4895-a712-b1efba58b250]** (SUPPORTED): The retrieved corpus contains the text of Section 235 of the Bharatiya Nyaya Sanhita, 2023.
- **Evidence (Chunk fab9445c-4a40-497e-b275-0695144092ca):** 1]

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- **[d9efba9f-b39d-4468-aa43-37e23d35c1b2]** (SUPPORTED): The provisions of Section 235 of the Bharatiya Nyaya Sanhita, 2023.
- **Evidence (Chunk fab9445c-4a40-497e-b275-0695144092ca):** 1]

- **[2a592573-6659-44ad-a08c-4a7fcf7e1938]** (SUPPORTED): The provisions of Section 235 of the Bharatiya Nyaya Sanhita, 2023
- *Evidence (Chunk fab9445c-4a40-497e-b275-0695144092ca):* 1]

- **[6b8b4990-2813-490a-b668-18afda0513a7]** (SUPPORTED): The retrieved corpus does not include any judicial judgments that
- *Evidence (Chunk fab9445c-4a40-497e-b275-0695144092ca):* 1]

- **[3fe526fb-741c-4ee2-86f3-1803037c6b86]** (SUPPORTED): No judgments from the retrieved corpus can be identified as apply
- *Evidence (Chunk fab9445c-4a40-497e-b275-0695144092ca):* 1]

7. Root Cause Analysis

- **Primary Cause:** UNKNOWN
- **Confidence:** 0.93
- **Reasoning:**
 - Starting root cause analysis for trace 8eb23758-4b1e-41d3-810c-2367ffbbdec56.
 - CORPUS passed (confidence=0.90).
 - RETRIEVER passed (confidence=0.95).
 - Skipping CHUNKING: no observable state in matrix.
 - GENERATOR passed (confidence=0.95).
 - GROUNDING passed (confidence=0.90).
 - No failures detected. Pipeline is healthy or all evidence is UNKNOWN.

8. Corrective Action Plan

- **[INFORMATIONAL] Review Retrieval Top-K / Reranker Cutoff**
 - Description: Retrieval efficiency is low (17%); consider reducing top-k or tightening the reranker score cutoff.
 - Expected Improvement: Reduced prompt size and lower latency/cost with no loss of verified support. (Metric: Chunk utilization)

9. Evidence Traceability

- **Primary Cause:** UNKNOWN
- **Derived From:** RootCauseAnalysis
- **Pipeline Stage:** UNKNOWN
- **Supporting Claims:** 22a6bbe7-9c32-4895-a712-b1efba58b250, d9efba9f-b39d-4468-aa43-37e23d35c1b2, 2a592573-6659-44ad-a08c-4a7fcf7e1938
- **Supporting Chunks:** fab9445c-4a40-497e-b275-0695144092ca, fab9445c-4a40-497e-b275-0695144092ca, fab9445c-4a40-497e-b275-0695144092ca

10. Overall Assessment

- **Major Strength:** RETRIEVER is healthy (confidence=95%): At least one retrieved chunk provides supporting evidence for
- **Major Weakness:** UNKNOWN
- **Next Priority:** Review Retrieval Top-K / Reranker Cutoff
- **Recommendation:** Pipeline is healthy. Monitor for distribution shift.

11. Framework Footer

- **Generated By:** X-RAG Diagnostic Framework
- **Artifact Version:** 1.0

APPENDIX

Artifacts Generated

✓ RAGTrace

artifacts/rag_traces/8eb23758-4b1e-41d3-810c-2367ffbdec56.json

✓ ClaimSet

artifacts/claim_sets/8eb23758-4b1e-41d3-810c-2367ffbdec56.json

✓ VerificationResults

artifacts/verification/8eb23758-4b1e-41d3-810c-2367ffbdec56.json

✓ PipelineStateMatrix

artifacts/pipeline_state_matrix/8eb23758-4b1e-41d3-810c-2367ffbdec56.json

✓ RootCauseAnalysis

artifacts/root_cause_analysis/8eb23758-4b1e-41d3-810c-2367ffbdec56.json

✓ CorrectiveActionPlan

artifacts/corrective_action_plan/8eb23758-4b1e-41d3-810c-2367ffbdec56.json

✓ DiagnosticEvaluationReport

artifacts/reports/8eb23758-4b1e-41d3-810c-2367ffbdec56.json